Alphafold 3 - Seminar Talk

- They simplified noise + no clash loss, so unique structures don't have any
  - ↳ the diffusion model was trained on AlphaFold 2 models (some Hallucination)
  - ↳ Not entirely fixed yet
- MSA is only 4 blocks with the new MSA module and then pairwise attention
- AlphaFold 3 is a much more light weight model (Longer sequences are possible)
- Pair-weighted averaging → (lightweight version of original attention)

(John Walker) - For antibodies multiple runs of AlphaFold is needed to get the highest confidence
↳ AF sample

- Diffusion model → make with constraints from pair repr. and input (Gradient descent)
- AlphaFold 3 → 20 jobs per day (but you can make extra accounts)
- 5000 amino acids total (5000 atoms)
- For non-commercial use only
  - ↳ May have changed
- Lipids can stabilize some structures and also create false binding
- Multiple cells may superimpose
- Covariation is computed in the even predict head
  - ↳ It's computed by MSA and
  - $w = (1-\lambda) \cdot \mathrm{inv}(\mathrm{cov}(x + \lambda I))$
  - ↳ Direct Coupling Analysis, you need to renormalize the weights
- Weights are "reduced" to a contact map
- $C = \mathrm{cov}(x)$
- val, vec = eigh(c)
- C = val * vec @ vec.T
- C^{-1} = (vec @ vec.T)/val
- inverse - downweighting the largest eigenvalue of the covariance matrix
- Eigendecomposition can be computed via power iteration
- Dominant eigen-vector can be approximated via random matrix multiplication

vec ~ C @ random

Code:

```
rep = msa.reshape(N, -1)
rep = rep - rep.mean(0)
```

```
for i in range(100): # OPM
    pair = (rep.T @ rep)/rep.shape[0]
```

```
# random weights
params = np.random.normal(size = (rep.shape[0], 1))
params = pair @ params
params /= np.sqrt(np.square(params).sum())
(More code)...
```

20 steps needed